

Learning the Price Response of Spatiotemporal EV Charging Flow: A Graph-Attentive Surrogate Model

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Abstract—The rapid growth of electric vehicles (EVs) and widespread deployment of fast charging stations (FCSs) greatly strengthened the coupling between coupled transportation and power distribution networks (TN-PDN). However, uncoordinated charging behaviors significantly pose operational challenges to PDNs, particularly as regular power demands and EV charging demands peak simultaneously. As such, the approach to leverage EV flexibility through a spatiotemporal pricing mechanism to guide orderly charging is of growing interest. The present work addresses this issue by proposing a deep-learning graph-attentive surrogate model to quantify the flexibility of EV charging demands by learning the nonlinear relationships between charging prices and EV charging responses. Specifically, our model seeks to approximate user equilibrium solutions in multitemporal optimal traffic–power flows by integrating a graph convolutional network (GCN) to encode the spatial topology of transportation networks (TNs), a bidirectional cross-attention mechanism to fuse global-topological features, and a transformer architecture to capture temporal interactions between traffic flow and environmental parameters. Simulation results validate the high accuracy of the proposed model, yielding an average error of 2.26%. Moreover, the framework effectively exploits EV flexibility to reduce system operating costs by 3.5%, while alleviating traffic congestion and improving the operational security of the PDN.

Index Terms—Coupled traffic–power networks, deep learning (DL), surrogate modeling, traffic assignment problem.

NOMENCLATURE

$E(L)$	Set of lines in a power distribution network (PDN).
$E(N)$	Set of buses in the PDN.
$C(i)$	Set of electrified roads connected to bus $i \in E(N)$.
$\pi(i)$	Child buses of bus $i \in E(N)$.
T_A	Set of links in a transportation network (TN).
T_A^{rg}	Set of regular links in a TN.
T_A^{ch}	Set of charging links in a TN.
T_N	Set of nodes in a TN.
$T(R)$	Set of origin nodes, $T(R) \in T(N)$.

$T(S)$	Set of destination nodes, $T(S) \in T(N)$.
$T(RS)$	Set of origin–destination (O-D) pairs.
$T(T)$	Set of time periods.
K_{rs}^e	Set of paths connecting an O-D pair $r-s$, where $r \in T(R)$, $s \in T(S)$, and $rs \in T(RS)$.
a_i and b_i	Energy production cost coefficients at bus i .
C_t	Electricity price at the main grid.
c_a^{ch}	Traffic flow capacity of charging link $a \in T_A^{\text{ch}}$.
c_a^{rg}	Traffic flow capacity of regular link $a \in T_A^{\text{rg}}$.
i_l^r	Square of current flow capacity of line $l \in E(L)$.
Δt	Duration of each time period $t \in T(T)$.
p_i^f	Upper active generation limit at bus i .
p_i^r	Lower active generation limit at bus i .
p_i^{ramp}	Generator ramp limits at bus i .
$p_{j,t}^{\text{reg}}$	Regular power demand at bus j in time period t .
$q_{i,t}^d$	Reactive power demand at bus i in time period t .
q_i^f	Upper reactive generation limit at bus i .
q_i^r	Lower reactive generation limit at bus i .
$q_{rs,t}^e$	Traffic demand between O-D pair $r-s$.
r_{ij}^l	Resistance of line l connecting buses i and j .
t_a^0	Free travel time on regular link $a \in T_A^{\text{rg}}$.
t_a^{co}	Free travel time on charging link $a \in T_A^{\text{ch}}$.
$U_{i,t}^f/U_{i,t}^r$	Bounds of squared voltage magnitude at bus i .
$V_{i,t}^f/V_{i,t}^r$	Bounds of voltage magnitude at bus i .
x_{ij}^l	Reactance of line l connecting buses i and j .
z_{ij}^l	Impedance of line l connecting buses i and j .
$\delta_{rs}^{a,k}$	Binary term, where $\delta_{rs}^{a,k} = 1$ if path k passes link a , and $\delta_{rs}^{a,k} = 0$ otherwise.
ω	Monetary cost () of travel time.
Ee	Charging demand of unit traffic flow.
λ_a^e	Base price (/kWh) at charging stations.
$c_{rs,k,t}^e$	Travel cost on path k between O-D pair $r-s$.
$f_{rs,k,t}^e$	Travel flow on path k between O-D pair $r-s$.
$\text{Fee}_{a,t}^e$	Charging service fee on link $a \in T_A^{\text{ch}}$.
$i_{ij,t}^l$	Square of current magnitude in line l from i to j .
$p_{i,t}^d$	Total active power demand at bus i .
$p_{i,t}^g$	Active power generated at bus i .
$p_{ij,t}^l$	Active power flow in line l from buses i to j .
$q_{i,t}^g$	Reactive power generated at bus i .
$q_{rs,t}^{e,\text{mod}}$	Modified traffic demand between O-D pair $r-s$.
$Q_{ij,t}^l$	Reactive power flow in line l from buses i to j .

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$r_{rs,t}^e$	Residual flow between O - D pair $r-s$.
$t_{a,t}^{rg}$	Travel time on link $a \in T_A^{rg}$.
$t_{a,t}^{ch}$	Average time spent by EVs on link $a \in T_A^{ch}$.
$t_{rs,k,t}^e$	Travel time on path k between O - D pair $r-s$.
$Toll_{a,t}$	Congestion toll on link $a \in T_A^{rg}$.
$u_{rs,t}^e$	Minimal travel cost between O - D pair $r-s$.
$U_{i,t}$	Square of voltage magnitude at bus i .
$x_{a,t}$	Aggregated traffic flow on link a .

I. INTRODUCTION

A. Motivation

SALES of electric vehicles (EVs) have increased explosively in recent years owing to increasing interest in low-carbon emission transportation options. According to the International Energy Agency, global EV ownership will exceed 240 million vehicles by 2030 [1]. Meanwhile, the rapid development of high-power and ultrafast EV charging technologies contributes toward continuous increases in the spatiotemporal variability and peak loads of charging demands. Moreover, both of these developments have led to deep coupling between TNs and PDNs, which makes their coordinated operation critical for ensuring overall system reliability and efficiency [2], [3].

The key to achieving the coordinated operation of coupled TN-PDN systems lies in the application of an effective spatiotemporal pricing mechanism for directing the charging operations of EVs. Hence, understanding how the spatiotemporal distribution of charging load responds to the dynamic pricing of charging stations is crucial [4]. Then, pricing mechanisms that capture this price-response relationship can be designed to reduce the operational challenges of PDNs for meeting both regular power demands and EV charging demands [5].

B. Literature Review

Issues affecting the coordination of coupled TN-PDN systems have been the subject of extensive research. For example, Sheng et al. [6] evaluated the security threats and network dependencies that can arise under close system coupling. Lu et al. [7] proposed an approach for measuring the dependence between coupled TNs and PDNs based on supply and demand, and Zhang et al. [8] analyzed the impact of increasing EV penetration on these dependencies. Zhao et al. [9] incorporated TN-PDN coupling within an optimal traffic-power flow model. Lv et al. [10] established a tri-level market mechanism framework for coordinating coupled systems to meet social optima. Xia et al. [11] further introduced a market mechanism based on a game process operating between the charging station operator and the EV user as two independent stakeholders. Li et al. [12] sought to coordinate TN-PDN system operation by adjusting EV charging behaviors using multiple scheduling methods of road tolls, charging fees, and plug-in fees. Li et al. [13] sought to optimize pricing and charge scheduling strategies for fast charging stations (FCSs) by applying a deep reinforcement learning (RL) approach for maximizing a robust reward function. Similarly, Qiu et al. [14] proposed a graph RL-based approach for optimizing the

TABLE I
COMPARISON OF MODELS IN RELEVANT LITERATURES

Reference	Methodology	Function	Coupling	Privacy	Price Response
[17]	EGCN-DBN	Surrogate	✓	Low	×
[18]	DNN	Surrogate	✓	Low	×
[19]	GCN	Prediction	×	-	×
[20]	DNN	Surrogate	✓	Medium	✓
[21]	STMGCN	Prediction	×	-	×
[24]	GCN-RNN	Prediction	×	-	×
Proposed	GCN-BCA-Tra	Surrogate	✓	High	✓

carbon-aware routing and scheduling of EVs within coupled power-transport networks. Wang et al. [15] introduced the concept of Age of Information into a synchronous log-linear learning framework for EV aggregators and ensured convergence to a welfare-maximizing Nash equilibrium. Wang et al. [16] established a two-stage TSO-DSO services provision framework that leverages a multiagent RL to coordinate EV active/reactive dispatch for frequency response and voltage regulation.

However, conventional solver-based approaches for coupled TN-PDN systems often suffer from limited computational efficiency and poor adaptability across dynamic multisenario environments. Moreover, the involvement of multiple stakeholders across transportation and power domains raises concerns over information privacy. These issues have generated increasing interest in the application of data-driven artificial intelligence methods, particularly deep learning (DL), owing to their strong capabilities in automatic feature extraction, accurate regression, and generalization, which enable efficient modeling and decision support under uncertainty. For example, Yuan et al. [17] proposed a novel alternative DL modeling approach for solving the stochastic traffic user equilibrium problem in a coupled TN-PDN system. Liang et al. [18] avoided the need for path generation iterations by training a neural network to identify optimal feasible paths for EVs. Wang et al. [19] proposed a heterogeneous spatiotemporal graph convolutional network (GCN) to predict EV charging demands. Chen et al. [20] proposed a DL-assisted collaborative optimization framework to achieve network decoupling and independent scheduling by predicting coupling variables through deep neural networks. Pourvaziri et al. [21] proposed a hybrid solution combining DL and queuing theory to solve the location-capacity problem of charging stations. Shi et al. [22] proposed a load forecasting method for multiple charging stations that can effectively share spatiotemporal relationships among stations. Al-Dahabreh et al. [23] applied a seasonal autoregressive integrated moving average with exogenous regressors (SARIMAXs) model to make informed optimal EV charging infrastructure expansions capable of coping reliably with future charging demands. Kim and Kim [24] utilized a spatiotemporal graph neural network (GNN) to mine potential spatiotemporal correlations between the EV charging loads of different charging stations from their charging load profile data. Shang et al. [25] improved PDN security and stability by developing a vehicle-to-grid (V2G) scheduling strategy based on a long short-term memory (LSTM) network. Table I

presents a comprehensive comparison between the proposed method and the existing methodologies regarding key technical features.

Nonetheless, applications of DL in the domain of coupled TN-PDN systems are substantially limited because the majority of extant approaches restrict prediction efforts to either the TN or PDN side exclusively. For example, existing studies have focused on the modeling of traffic flow characteristics, such as traffic flow distributions [17] and the travel route selections of vehicle users [18]. As a result, the modeling and optimization of important mechanisms for guiding EV user behaviors, such as charging prices and service fees, remain inadequate for addressing the demands of optimal scheduling within coupled TN-PDN systems.

C. Contributions

The present work addresses this issue by proposing the DL-based surrogate modeling approach illustrated in Fig. 1, which takes environmental parameters and traffic conditions as inputs, and outputs spatiotemporal charging load patterns for FCSs under price signals. Accordingly, the proposed approach enables fast and accurate analysis of the price flexibility of spatiotemporal charging loads, and thereby facilitates the coordinated and coupled operation of TN-PDN systems. The detailed contributions of the current work are specified as follows.

- 1) We develop a learning approach to model the price response of spatial-temporal EV charging flows, incorporating an MTOTPF model managed by an independent system operator (ISO) and optimized through charging service charge scheduling to achieve optimal social benefits.
- 2) We propose a graph-attentive surrogate model based on a GCN-BCA-Transformer architecture to capture the spatiotemporal dynamics and price dependencies of EV charging flows. The model represents an extended TN with charging stations as a graph, using a GCN to extract link-level spatial features, a bidirectional cross-attention mechanism to enable interaction between global OD demand and local network topology, and a Transformer with learnable positional encoding to capture the spatiotemporal dynamics of traffic flow.
- 3) Numerical results verify that the proposed surrogate modeling method based on GCN-BCA-Transformer network outperforms existing DL-based supervised learning methods. We also investigate the spatiotemporal impact of scheduling means on the coordinated operation of the coupled PDN-TN system.

The rest of the article is structured as follows. Section II provides the nonconvex MTOTPF model. Section III elaborates on the agent model framework based on the GCN-BCA-Transformer architecture. Section IV summarizes the numerical computational results, and Section V concludes this article.

II. MODEL FORMULATION

The coupled PDN and TN framework considered in the present study is illustrated in Fig. 2, where we assume the

presence of a nonprofit ISO to manage these two networks with the aim of achieving maximum social benefits. To this end, the ISO is qualified to set dispatching fees, such as charging service fees (CSFs), and dispatching local generator units. Here, EV users determine their departure times and travel routes in response to traffic conditions and the CSFs set by the ISO. Generators at each PDN node adjust their power outputs according to the resulting charging loads, and formulate flexible demand schedules. The ultimate goal is to maximize overall social benefit by minimizing both travel costs and energy costs. The feasibility and tractability of the proposed coupled system model are ensured by applying the following key assumptions.

- 1) All EVs requiring charging services are assumed to have the same initial state of charge (SOC) when departing from their origin nodes.
- 2) Each EV selects the route that minimizes its total travel cost, which consists of both travel time cost and charging cost. Each selected path must contain only a single FCS.
- 3) Static TN and PDN topologies are assumed throughout the period of operation.

A. TN Model

The TN is represented by the connected graph $G_T = [T_N, T_A]$, where T_N is the set of origin, destination, and intersection nodes, and T_A represents the set of links corresponding to lanes or sections of road. Each EV in the TN starts from an origin node $r \in T(R)$ and travels to a destination node $s \in T(S)$. The traffic sections are expanded into three categories as proposed previously [26], [27], which include regular links $a \in T_A^{\text{rg}}$ with no FCSs, EV charging links $a \in T_A^{\text{ch}}$ that contain an FCS, and bypass links $a \in T_A^{\text{by}}$ that are associated with EVs passing by FCSs without charging. It is assumed that each EV selects a path that includes at least one charging link. Based on this formalism, the semidynamic traffic assignment problem can be expressed as follows:

$$t_{a,t}^{\text{rg}}(x_{a,t}) = t_a^0 \left[1 + 0.15 \left(\frac{x_{a,t}^{\text{rg}}}{C_a^{\text{rg}}} \right)^4 \right], \quad a \in T_A^{\text{rg}} \quad (1)$$

$$t_{a,t}^{\text{ch}}(x_{a,t}) = t_a^{c0} \left[1 + J \left(\frac{x_{a,t}^{\text{ch}}}{C_a^{\text{ch}} - x_{a,t}^{\text{ch}}} \right) \right], \quad a \in T_A^{\text{ch}} \quad (2)$$

$$t_{rs,k,t}^e = \sum_a t_{a,t}(x_{a,t}) \delta_{rs}^{a,k} \quad (3)$$

$$r_{rs,t}^e = \sum_{k \in K_{rs}^e} \frac{f_{rs,k,t}^e t_{rs,k,t}^e}{\Delta t} \quad (4)$$

$$q_{rs,t}^{e,\text{mod}} = q_{rs,t}^e + \frac{1}{2} r_{rs,t-1}^e - \frac{1}{2} r_{rs,t}^e \quad (5)$$

$$\sum_{k \in K_{rs}^e} f_{rs,k,t}^e = q_{rs,t}^{e,\text{mod}} \quad (6)$$

$$x_{a,t}^{\text{rg}} = \sum_{rs} \sum_{k \in K_{rs}^e} f_{rs,k,t}^e \delta_{rs}^{a,k} \quad (7)$$

$$x_{a,t}^{\text{ch}} = \sum_{rs} \sum_{k \in K_{rs}^e} f_{k,rs,t}^e \delta_{rs}^{a,k}, \quad a \in T_A^{\text{ch}} \quad (8)$$

$$\begin{aligned}
c_{rs,k,t}^e &= \sum_{a \in T_a^{\text{rg}}} (\omega t_{a,t}^{\text{rg}} + \text{Toll}_{a,t}) \delta_{rs}^{a,k} \\
&+ \sum_{a \in T_a^{\text{ch}}} (\omega t_{a,t}^{\text{ch}} + \lambda_a^e E_e + \text{Fee}_{a,t}^e) \delta_{rs}^{a,k} \quad (9) \\
\begin{cases} f_{rs,k,t}^e \geq 0 \\ c_{rs,k,t}^e - u_{rs,t}^e \geq 0 \\ f_{rs,k,t}^e (c_{rs,k,t}^e - u_{rs,t}^e) = 0 \end{cases} &\quad \forall k \in K_{rs}^e \quad \forall rs \quad \forall t. \quad (10)
\end{aligned}$$

Equation (1) represents the BPR function, which is generally used to describe the relationship between the travel time $t_{a,t}^{\text{rg}}$ and the traffic flow $x_{a,t}^{\text{rg}}$ of a regular link $a \in T_A^{\text{rg}}$. For regular links, it is generally not mandatory for $x_{a,t}^{\text{rg}}$ to be less than the link capacity c_a^{rg} , although FCSs at charging links must meet this condition. Equation (2) adopts the Davidson function of queuing theory to represent the queuing time of EVs at FCSs, where J is typically set to be 0.05. Equation (3) represents the travel time on path k . An indicator variable is introduced to represent the relationship between link a and path k in period t , where the variable is taken as 1 if the road section belongs to the path, and is otherwise taken as 0. Equation (4) represents the residual flow of path k in period t . Assuming that vehicles leave uniformly in each period, the departure rate of path k is expressed as $\sum_{k \in K_{rs}^e} f_{rs,k,t}^e / \Delta t$, and the residual flow on path k can be expressed as $\sum_{k \in K_{rs}^e} f_{rs,k,t}^e t_{rs,k,t}^e / \Delta t$. Equation (5) represents the modified demand of each $O-D$ pair in period t . Generally, this is defined as $q_{rs,t}^{\text{mod}} = q_{rs,t}^e + \alpha r_{rs,t-1}^e - \alpha r_{rs,t}^e$, and α is set to 1/2 in this article for simplicity. Accordingly, the modified demand includes half of the residual flow in the previous period, and half of the residual flow is subtracted from the current traffic demand. Equation (6) indicates that the traffic flow on each link is equal to the sum of the flows on all paths passing through that link. Equations (7) and (8), respectively, represent the flow of EVs on regular links and charging links. Equation (9) represents the travel cost of an EV on path k , which includes the time cost and the charging fee at the corresponding FCS. Equation (10) represents the complementary condition of the Wardrop-UE principle.

B. PDN Model

As illustrated in Fig. 2, all FCSs in the TN are served by the corresponding PDN, which is usually a radial network that is generally represented by a tree topology. This article adopts the following SOC-based DistFlow model to describe the power flow distribution in the PDN [28]:

$$P_{ij,t}^l + P_{j,t}^g - r_{ij}^l i_{ij,t}^l = \sum_{k \in \pi(j)} P_{jk,t}^l + P_{j,t}^d \quad \forall l, t \quad (11)$$

$$Q_{ij,t}^l + Q_{j,t}^g - x_{ij}^l i_{ij,t}^l = \sum_{k \in \pi(j)} Q_{jk,t}^l + Q_{j,t}^d \quad \forall l, t \quad (12)$$

$$U_{j,t} = U_{i,t} - 2(r_{ij}^l P_{ij,t}^l + x_{ij}^l Q_{ij,t}^l) + (z_{ij}^l)^2 i_{ij,t}^l \quad \forall l, t \quad (13)$$

$$i_{ij,t}^l \geq \frac{(P_{ij,t}^l)^2 + (Q_{ij,t}^l)^2}{U_{i,t}} \quad (14)$$

$$i_{ij,t}^l \leq i_l^r \quad \forall l, t, U_{i,t} \leq U_{i,t} \leq U_{i,t}^r \quad \forall i, t \quad (15)$$

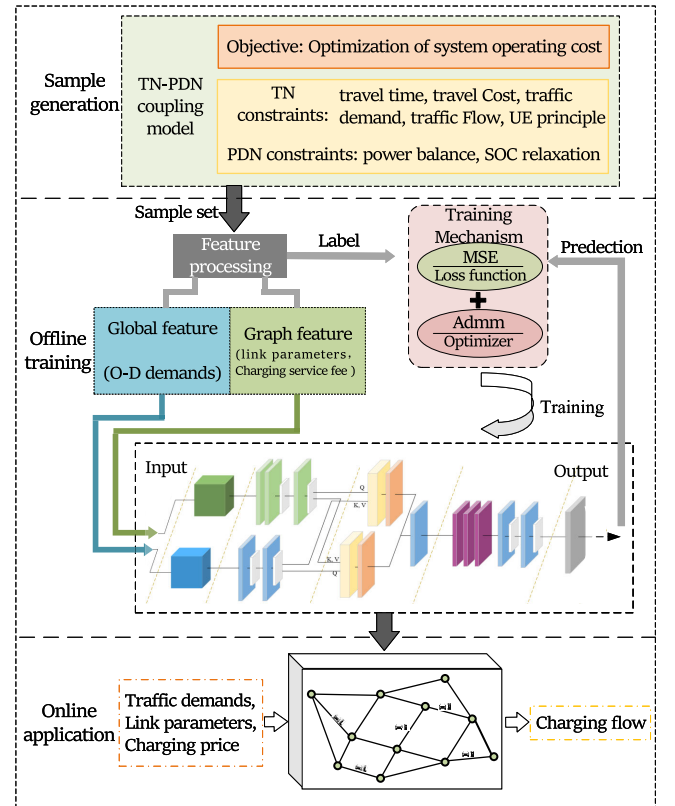


Fig. 1. Framework of the DL-based surrogate modeling approach.

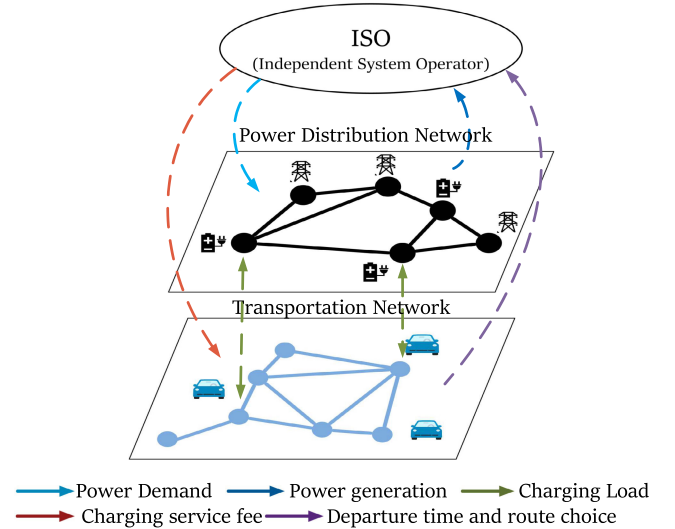


Fig. 2. Coupled PDN and TN.

$$p_i^f \leq p_{i,t}^g \leq p_i^r, q_i^f \leq q_{i,t}^g \leq q_i^r \quad \forall i \quad (16)$$

$$-p_i^{\text{ramp}} \leq p_{i,t}^g - p_{i,t-1}^g \leq p_i^{\text{ramp}} \quad (17)$$

$$p_{j,t}^d = p_{j,t}^{\text{reg}} + \sum_{a \in C_e(j)} x_{a,t}^{\text{ch}} E_e, \quad a \in C_e(j). \quad (18)$$

Equations (11) and (12) are the node active and reactive power balance equations, respectively. Equation (13) defines the relationship between the line voltages and the currents. Equation (14) is the second-order cone relaxation constraint on line currents, and (15) specifies the bounds on the line currents and node voltages. Equation (16) stipulates the upper/lower

limits of distributed generation (DG) output. Equation (17) formulates the DG ramp limits. This constraint mathematically links adjacent time periods by restricting the maximum allowable power variation. The nodal demands of the PDN are given in (18), where each charging station in the TN is assumed to be served solely by an electrical bus in the PDN, and the energy consumed by each charging station is proportional to the corresponding link traffic flow.

C. Multitemporal Optimal Traffic and Power Flow Model

The goal of the ISO is to minimize both the travel costs and energy costs of the system. Accordingly, the objective function is expressed with four individual objectives as follows:

$$\begin{aligned} & \sum_{t \in T(T)} C_t P_{\text{sub},t} + \sum_{t \in T(T)} \sum_{i \in E(N)} \left[a_i (p_{i,t}^g)^2 + b_i p_{i,t}^g \right] \\ & + \sum_{t \in T(T)} \sum_{a \in T_A^{\text{rg}}} (\omega t_{a,t}^{\text{rg}} + \text{Toll}_{a,t}) x_{a,t}^{\text{rg}} \\ & + \sum_{t \in T(T)} \sum_{a \in T_A^{\text{ch}}} (\omega t_{a,t}^{\text{ch}} + \lambda_a^e E_e + \text{Fee}_{a,t}^e) x_{a,t}^{\text{ch}} \end{aligned} \quad (19)$$

where the first two objectives represent the operating cost of the PDN, which is composed of the power generation cost and the cost of purchasing power from the main grid. The second two objectives, respectively, represent the travel time costs and charging costs of all EV users in the TN. According to the Wardrop-UE principle, $O-D$ should have the same minimum travel cost for all the selected paths between $r - s$. This yields the transformation of the second term below the objective function in (19) [29]. This yields the following simplified objective function:

$$\begin{aligned} F_{\text{obj}} = & \sum_{t \in T(T)} C_t P_{\text{sub},t} + \sum_{t \in T(T)} \sum_{i \in E(N)} \left[a_i (p_{i,t}^g)^2 + b_i p_{i,t}^g \right] \\ & + \sum_{t \in T(T)} \sum_{rs \in T(\text{RS})} u_{rs,k,t}^e q_{rs,t}^{e,\text{mod}}. \end{aligned} \quad (20)$$

It is noted that solving the coordinated model (1)–(18) and (20) faces two primary challenges. First, this model is a nonlinear programming (NLP) problem and computationally intensive, making it difficult to support real-time decision-making. Second, the power and transportation systems involve multiple stakeholders, and coordinated dispatch requires repeated exchange of sensitive information, raising privacy concerns. To address these issues, this article proposes a learning-based approach to efficiently capture the mapping relationship among electricity prices, traffic flows, and charging demands. This approach enables effective real-time coordination between TN and PDN while protecting information privacy.

III. SOLUTION FRAMEWORK

The proposed DL approach essentially learns the optimal solutions of the NLP model (i.e., the spatiotemporal charging load responses under varying electricity prices and traffic flow conditions). We assume the presence of a centralized coordinator or regulator that has access to the parameter information of PDN, TN, and EVs. This entity solves the NLP model under

numerous scenarios to generate extensive labeled samples for training the DL model. During the online application, the trained DL model only requires real-time electricity price and traffic flow information to rapidly predict the corresponding charging load response, thereby supporting effective and real-time coordination between power and transportation systems.

To construct an effective data-driven DL modeling framework that can be applied to traffic networks in different uncertain scenarios, we have established two core design goals. First, the model needs to be capable of effectively capturing the input graph features (e.g., link parameters) and global features (e.g., traffic demand). Second, after extracting these features, the model can be accurately mapped to the charging demands of EVs and has good generalization ability. Therefore, a hybrid GCN–BCA–transformer network is proposed, and its architecture is illustrated in Fig. 3.

A. Graph Convolutional Network

In real-world scenarios, many datasets are composed of non-Euclidean data that lack a fixed temporal or spatial order and exhibit irregular structures. Traffic networks are a typical example, with complex spatial correlations that conventional convolutional neural networks (CNNs) struggle to capture. Here, GCNs address this issue by extending convolution operations to graph-structured data, and accordingly offer clear advantages over CNNs for predicting EV traffic flows [30].

A GCN describes the structure of a graph in terms of an adjacency matrix A . Conventionally, TNs are modeled with intersections as nodes and roads as edges. However, this conventional modeling approach is not appropriate in the present work because the applied traffic model adopts charging links and bypath links. Therefore, links are regarded herein as nodes of the graph and the connection relationship between links is regarded as edges of the graph, where a charging link is regarded as having an undirected edge connection with the link where the FCS is located. Therefore, the adjacency matrix is defined as follows:

$$A_{i,j} = \begin{cases} 1, & \text{adjacent} \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

A GCN interacts with node information and extracts features along the edges of the graph without destroying the graph structure of the data. This gives graph convolutional neurons unique advantages when dealing with graph-structured data. The core formula of GCN is as follows:

$$H^{(l+1)} = \sigma \left(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right) \quad (22)$$

where $\sigma()$ is the nonlinear activation function. In this study, we adopt the ReLU function [31]. $\hat{D}$ is the degree matrix defining the number of edges terminating at each node, where the diagonal elements are the degrees representing the number of neighbors of the nodes, and are used to normalize the adjacency matrix; $\hat{A} = A + I$ is the adjacency matrix plus the identity matrix, which represents the relationship between the node and itself; $H^{(l)}$ constitutes the feature matrix of layer l ; and $W^{(l)}$ represents the weight matrix of layer l , which is used to perform linear transformation on the nodes.

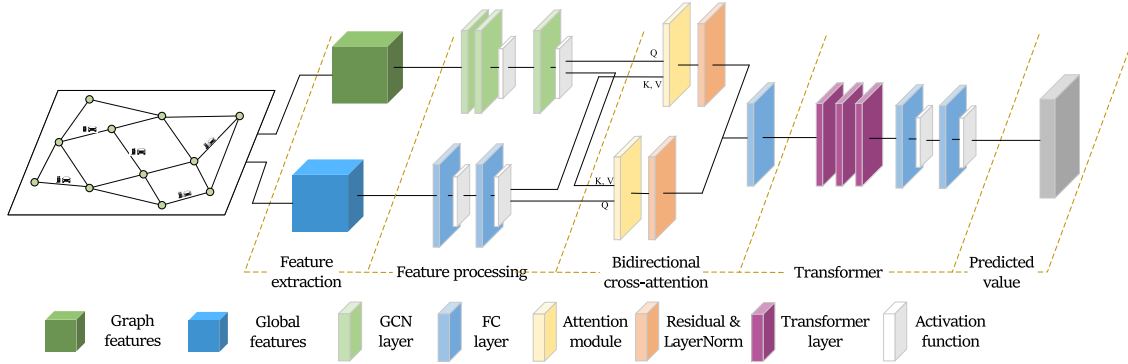


Fig. 3. Proposed neural network architecture of the graph-attentive surrogate model.

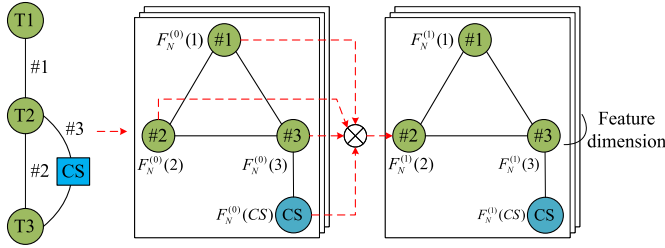


Fig. 4. Schematic of the extraction of spatial dependencies by a GCN.

This extraction process is illustrated in Fig. 4 for a three-node TN topology with a single CS as an example. Let $F_N^{(l)}(n)$ represent the feature matrix of the nodes. At the input layer, $F_N^{(0)}(n)$ is composed of the original features of the links, including the link capacity c_a , the travel time $t_{a,t}$, and the CSF $\text{Fee}_{a,t}^e$. After passing through each layer of the GCN, the link nodes will obtain the information of their adjacent links. As shown in Fig. 4, only two GCN layers are needed to broadcast the features of the charging stations to all nodes. For all nodes, the process of information acquisition is carried out independently and simultaneously, and the dimension of node features changes with changes in the layer index. Finally, the feature matrix of the link nodes is output for subsequent processing.

B. Bidirectional Cross Attention

As discussed above, the present work applies a BCA mechanism to fuse graph features with global features, including traffic demand between $O-D$ pairs, according to the scheme illustrated in Fig. 3. The BCA dynamically allocates attention weights in both forward and backward directions, enabling mutual guidance between global and graph features. As such, this approach preserves local spatial structure and global contextual relationships better than conventional one-way attention or simple concatenation mechanisms, and thereby yields a unified context-aware feature representation.

Specifically, $O-D$ traffic demand vectors are projected into the same dimension as GCN-generated graph node features. A BCA module then facilitates interaction between these features. This corresponds to two attention flows, including graph node features $\rightarrow$ global features (G2E) and global features $\rightarrow$ graph node features (E2G) interaction processes.

The G2E interaction process takes graph nodes as the Query (Q) terms and global features as the Key/Value (K and V) terms, and thereby captures the influence of external factors on graph node features according to the following operation:

$$\text{Attn}_{\text{G2E}} = \text{Softmax} \left(\frac{H_{\text{GCN}} W_Q (H_{\text{Extra}} W_K)^T}{\sqrt{d_k}} \right) H_{\text{Extra}} W_V. \quad (23)$$

Accordingly, the G2E interaction process enhances the overall perception of $O-D$ traffic demand by road section nodes. Similarly, the E2G interaction process takes global features as the Q terms and graph node features as the K and V terms, and thereby enhances the adaptability of external features to the graph structure according to the following operation:

$$\text{Attn}_{\text{E2G}} = \text{Softmax} \left(\frac{H_{\text{Extra}} W'_Q (H_{\text{GCN}} W'_K)^T}{\sqrt{d_k}} \right) H_{\text{Extra}} W'_V. \quad (24)$$

Accordingly, the E2G interaction process adapts $O-D$ traffic demand to the road network topology. The BCA mechanism automatically learns the correlation strength and dynamic weight distribution between TN node features and the global features through backpropagation (BP) optimization. The model captures both the local spatial dependencies of graph nodes and the global context of the TN, thereby synthesizing multiscale features to enhance information representation.

The attention outputs Attn_{G2E} and Attn_{E2G} are then added to their corresponding features with residual connections and LayerNorm to enhance representations

$$H_{\text{G2E}}^{\text{new}} = \text{LayerNorm} (H_{\text{G2E}} + \text{Dropout} (\text{Attn}_{\text{G2E}})) \quad (25)$$

$$H_{\text{E2G}}^{\text{new}} = \text{LayerNorm} (H_{\text{E2G}} + \text{Dropout} (\text{Attn}_{\text{E2G}})). \quad (26)$$

Finally, splicing is applied to obtain the final fused features that enable the transformer module to model high-order dependencies directly without relearning basic relationships among service fees, $O-D$ demands, and traffic flows.

C. Transformer

The transformer module applied in the present work applies an encoder core structure composed of multiple stacked basic modules, each of which contains a multihead attention layer, a feedforward neural network (FFN) layer, residual connections, and layer normalization, and replaces the decoder with an FC layer to output the charging station flow. The transformer

module thereby achieves end-to-end mapping from the fused features to the relationship matrix.

The self-attention operation is the core process of the transformer encoder, and the multihead attention mechanism can be regarded as attention integration, which captures the global dependencies between nodes through multiple parallel sets of linear projections and attention calculations. Specifically, the Q , K , and V are obtained by multiplying the input fused feature X by the corresponding weight matrices W^Q , W^K , and W^V as follows:

$$\begin{cases} Q = W^Q * X \\ K = W^K * X \\ V = W^V * X. \end{cases} \quad (27)$$

The magnitude of the attention weights is determined by calculating the similarity between Q and K as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (28)$$

where d_k is the dimension of the input K vector. Then, the correlation information between the input sequences is captured through each attention head

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (29)$$

where W_i^Q , W_i^K , and W_i^V , respectively, represent the weight matrices corresponding to Q , K , and V in the attention header. Finally, the outputs of the various attention heads are concatenated and projected via an FC layer

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_n)W^O \quad (30)$$

where n represents the number of stacked attention mechanisms and the $\text{Concat}()$ function represents dimension concatenation. Accordingly, the multihead process enables the transformer module to capture diverse subspace features.

However, the conventional sinusoidal positional encoding employed above may not be optimal because the model already incorporates a GCN. Thus, the fused feature input X is modified as follows based on a learnable positional encoding matrix P , which is initialized randomly and optimized via BP

$$X_{\text{encoded}} = X + P. \quad (31)$$

This approach demonstrates greater advantages in learning the weights reflecting the importance of charging links and regular links, which complements the local extraction ability of the GCN and the global attention of the transformer module.

The FFN consists of two linear layers with ReLU activation

$$F_{\text{FN}}(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2 \quad (32)$$

where x is the output vector obtained after normalizing the attention layer, W is the weight vector, and b is the bias term.

Meanwhile, residual connections and layer normalization are introduced behind both the multihead attention layer and the FFN to alleviate the vanishing gradient problem in the deep network, which facilitates the propagation of the fused features of the TN in the model. To enhance the model's generalization capability and mitigate overfitting, a dropout regularization strategy with a rate of 0.1 is employed.

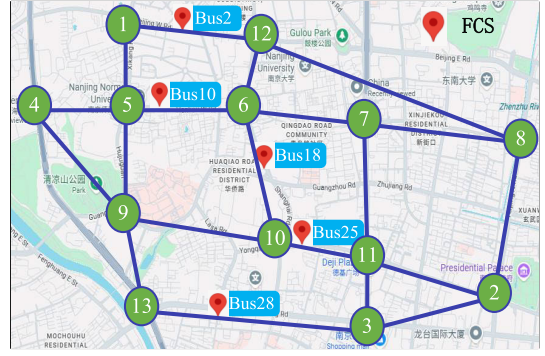


Fig. 5. Topology of the coupled TN covered with PDN system.

After performing position encoding, the FC layer decoder is used to generate the prediction matrix and adjust the output dimension because the prediction target of the model is the EV traffic flow of charging stations over multiple periods, rather than a variable-length sequence, such as text. Accordingly, the predicted traffic flow x_o is calculated as follows:

$$x_o = \text{FC}(x_c) \quad (33)$$

where x_c is the fused feature vector processed by the encoder.

The mean-squared error (mse) was applied as the objective function for optimal parameter estimation during model training according to the following formulation:

$$\text{loss} = \text{mse} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (34)$$

where n is the total number of predicted values, y_i is the truth EV traffic flow of charging stations, and $\hat{y}_i$ is the predicted traffic flow. In addition, Adam was utilized to minimize the loss iteratively through the gradient descent mechanism.

IV. CASE STUDIES

The topology of the coupled TN-PDN system employed in the case studies is presented in Fig. 5. The TN is represented by a modified 13-link Nguyen–Dupius network with parameter settings calibrated using map-measured data from Nanjing, China. The PDN is represented by a modified IEEE 33 bus network. The system also consists of five FCSs, where FCS1–FCS5 are connected to buses 2, 10, 18, 25, and 28 in the PDN, respectively. Here, link 5, at which FCS2 is located, has the shortest travel time among all the links with charging stations. Therefore, users are more inclined to charge at this station, resulting in congestion on links 5–6 and 6–7. For fixed parameters, we applied a monetary travel time cost of $\omega = 5\$/h$ and a fixed service charge of $\text{Fee}_{a,t}^e = 0.5\%$, which was allowed to be adjusted by 50% up or down dynamically. A proxy modeling method was developed based on the PyTorch 2.4.0 framework of Python 3.9.20, and executed on a laptop with an Intel¹ Core² i7-14700HX CPU and 16 GB of RAM. The original nonconvex NLP problem was solved using IPOPT

¹Registered trademark.

²Trademarked.

TABLE II
STATISTICAL ANALYSIS OF SOC RELAXATION GAP

Metric	Value (p.u.)
Maximum Gap	3.21×10^{-6}
Minimum Gap	1.87×10^{-7}
Average Gap	9.15×10^{-7}

TABLE III
HYPERPARAMETER SETTINGS OF THE PROPOSED MODEL

Module	Parameter	Value
GCN	Number of Layers	2
	Hidden Dimension	16
	Activation Function	ReLU
Transformer	Embedding Dimension	128
	Attention Heads	4
	Encoder Layers	3
	Dropout Rate	0.1
Training	Batch Size	32
	Learning Rate	1×10^{-3}
	Optimizer	Adam

3.11 in GAMS. All detailed parameters of the case study are available in the online supplement.³

The traffic flow data employed for model training and testing was obtained via simulations conducted based on traffic demands at different times of the day, with every five time periods taken as a group. Different scenarios were simulated by randomly changing some core parameters affecting the operation of the TN, such as *O-D* pair traffic demands and the free travel time of links. Then, we recorded the labels generated over each group of time periods during the simulation of each scenario, including the traffic flow of each FCS and the scheduled CSF. A total of 14 000 groups of samples were generated, and 80% of these were selected randomly as the training dataset and the remaining 20% was applied as the testing dataset.

To ensure the high fidelity of the training data, we validate the tightness of the SOC relaxation within the DistFlow model prior to training the surrogate model. The relaxation gap for line (i, j) at time t is quantified as $\epsilon_{ij,t} = |i_{ij,t}U_{i,t} - ((P_{ij,t})^2 + (Q_{ij,t})^2)|$. Table II summarizes the statistical distribution of these residuals across the entire dataset. The results indicate that the maximum and average gaps are limited to 3.21×10^{-6} p.u. and 9.15×10^{-7} p.u., respectively. Consequently, the relaxation-induced inconsistency is demonstrated to be negligible.

To ensure the reproducibility of the experiments, the detailed hyperparameter settings of the neural network architecture and the training strategies are summarized in Table III.

A. Model Accuracy and Computational Efficiency

The effectiveness of the proposed surrogate modeling method employing the GCN–BCA–transformer architecture

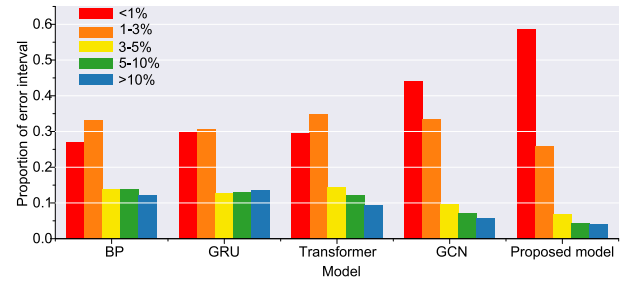


Fig. 6. Proportion of error intervals for different models.

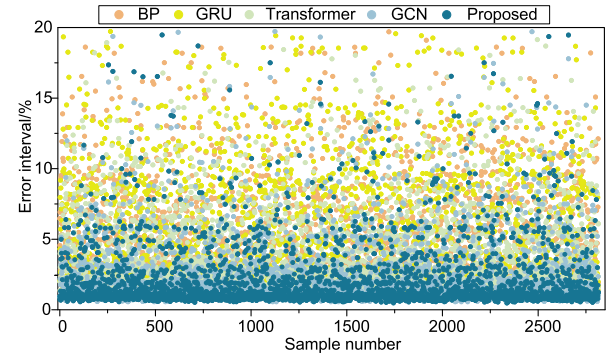


Fig. 7. Comparison of average prediction errors obtained by all models.

was evaluated by comparing its performance with that obtained using different existing DL-based methods, including a BP neural network, a gated recurrent unit (GRU) network, a GCN alone, and a transformer architecture alone. We first investigated the accuracy errors of the traffic flow at charging stations predicted by different methods in the test dataset compared with the true values, and the proportion of these errors was compared as shown in Fig. 6. In addition, the accuracy errors obtained by the different methods for each sample in the testing dataset are compared in Fig. 7.

As can be seen from Fig. 6, the proposed model provided the highest proportion of predictions with an error range of less than 1%, while also providing the lowest proportions of predictions with average error of 1%–3%, 3%–5%, 5%–10%, and >10%. Meanwhile, for models without using graph structure, due to the benefits brought by the self-focus mechanism in the transformer architecture, capturing remote dependencies, the proportion of average errors of 5%–10% and >10% is smaller compared to the prediction results obtained by BP and GRU methods. Applying the GCN alone provided the second-best performance; this marginal enhancement can be expected because GCN learns the spatial dependence of the TN by more effectively utilizing the topological structure of the TN to form an adjacency matrix.

Furthermore, as can be seen from Fig. 7, the proposed model maintains the lowest average error within the overall sample range, indicating that it has stronger robustness in different scenarios. Hence, the proposed model is clearly seen to benefit from the utilization of the topological structure of the TN by the GCN to model spatial dependences effectively, along with the capture of long-distance dependencies by the self-attention mechanism in the transformer architecture.

³Supplement available at: <https://github.com/han-jc/network-parameters>

TABLE IV
EVALUATION PARAMETERS FOR DIFFERENT MODELS
IN TESTING SAMPLES

	MSE	RMSE	MAE	R^2	Average error
BP	105.75	10.28	6.72	0.9817	5.27%
GRU	119.14	10.91	6.71	0.9792	5.56%
Transformer	73.10	8.55	5.16	0.9874	3.98%
GCN	50.66	7.1181	3.65	0.9913	2.84%
Proposed model	44.94	6.70	2.92	0.9922	2.26%

TABLE V
RUNTIME COMPARISON BETWEEN SOLVER AND SURROGATE MODEL

Method	Solution time	Relative speed-up
IPOPT Solver	73.8 s	1×
Proposed	35 ms	2108×

Finally, the prediction performances of the various methods are further clarified by the results in Table IV. The proposed model demonstrates superior performance under various evaluation parameter indicators. The average errors have increased by 57.93%, 59.35%, 43.22%, and 21.83%, respectively, compared with BP, GRU, Transformer, and GCN. This confirms that the proposed model has a high accuracy rate and excellent generalization ability in different TN uncertain scenarios.

The computational efficiency of the proposed model is verified by comparing its average execution time with that of traditional benchmarks. As shown in Table V, the complex nonconvex and high-dimensional nature of the TN-PDN problem renders traditional solvers computationally inefficient. In contrast, the proposed surrogate model reduces solution times to the millisecond level. This dramatic speedup is important for empowering the aggregation of charging stations into virtual power plants to participate in real-time flexible grid regulation [33].

B. Validation on Real-World Field Data

To further validate the practical applicability and generalization capability of the proposed surrogate model, we extended our experiments beyond simulation environments by utilizing a real-world dataset. We collected operational data from five public fast-charging stations located in Nanjing, China. The dataset is sourced from [32]. The proposed model was trained on this real-world dataset to predict the spatiotemporal charging loads.

Fig. 8 visualizes the load tracking performance over a 144-h horizon. The proposed model achieves high fidelity with an R^2 of 0.9411 and an RMSE of 749.28 kW. Although empirical load profiles are characterized by significant stochasticity and sharp peaks due to behavioral uncertainty, our approach robustly captures these complex temporal dynamics, including rapid ramp events during peak hours and fluctuations during valley periods. These results validate that the model has effectively mastered the nonlinear correlations between pricing signals and charging demand, establishing its reliability for online dispatch in the coupled TN-PDN system.

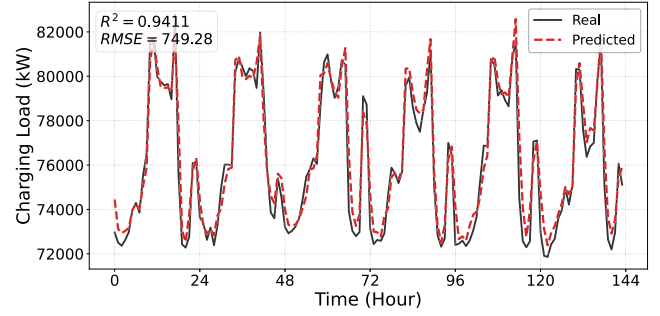


Fig. 8. Comparison between the real-world charging load and the predicted load by the proposed model over a 144-h period.

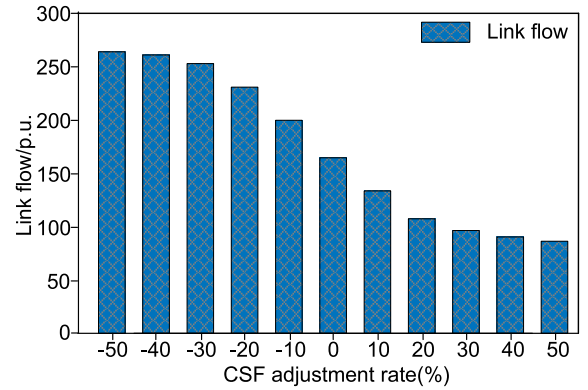


Fig. 9. Predicted vehicle flow at charging stations under different CSF adjustment rates.

C. Interpretation of Model Mechanism

To address the "opaque" nature of DL and validate that the proposed model successfully learns the underlying operational logic of the coupled system, we analyze the influence of key inputs on the outcomes.

Fig. 9 depicts the sensitivity of charging flow to CSF variations ($\pm 50\%$). The results reveal a distinct inverse monotonic trend, confirming that the surrogate model correctly encodes rational economic logic rather than functioning as an unconstrained opaque model. Furthermore, the observed saturation effect at higher prices is characterized by reduced marginal flow deviations. This aligns with realistic limits on user flexibility due to station availability and energy constraints.

Beyond pricing incentives, the model explicitly encodes inherent spatiotemporal dependencies. Spatially, the ablation study (see Table IV) reveals that excluding the GCN module incurs a 21.83% error penalty, underscoring the criticality of topological information. Temporally, the precise tracking of diurnal periodicities (see Fig. 8) validates the model's fidelity in capturing complex dynamic couplings. Collectively, these analyses confirm the physical consistency and interpretability of the proposed surrogate for coupled PDN-TN.

D. Flexibility Provided by Dynamic CSF for TN and PDN

The guiding effect of dynamic CSF scheduling on EVs was evaluated by comparing the EV traffic flows during time periods 16–20 (i.e., time period groups 1–5) under fixed and

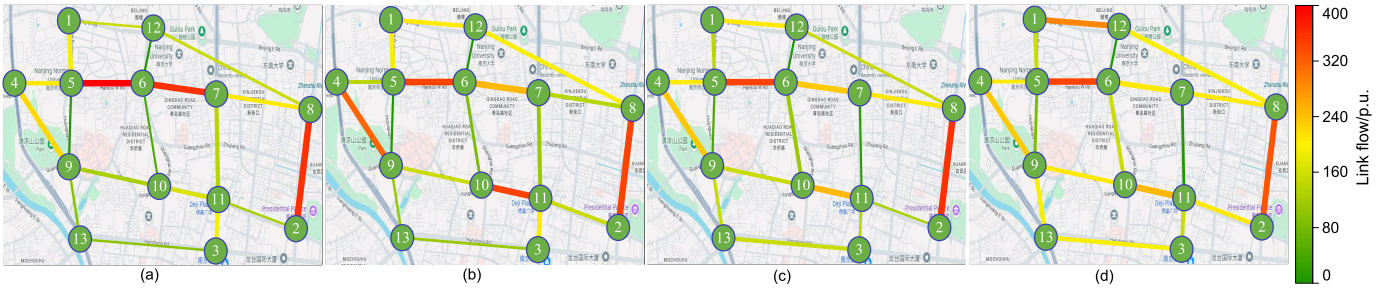


Fig. 10. EV traffic flows (p.u.) on each link under different CSF conditions. Fixed condition of (a) time period 1 and (b) time period 5. Dynamic condition of (c) time period 1 and (d) time period 5.

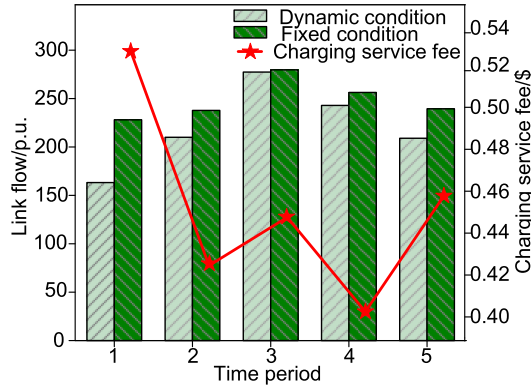


Fig. 11. Traffic flows at FCS2 obtained with and without dynamic CSF.

dynamic CSF conditions. The results obtained under the fixed condition during time group periods 1 and 5 are presented in Fig. 10(a) and (b), while those obtained under the dynamic condition during time group periods 1 and 5 are presented in Fig. 10(c) and (d), respectively.

As can be seen from Fig. 10(a) and (b), EV users are more inclined to choose links with the shortest travel times under fixed CSF conditions, which results in concentrated traffic and potential congestion. In contrast, we note from Fig. 10(c) and (d) obtained under dynamic CSF scheduling that a number of traffic sections are effectively diverted to underutilized sections, such as 1–12 and 10–11, and thereby alleviating traffic congestion. It can also be found that dynamic CSF scheduling generates a traffic flow distribution that is more balanced over time. These results confirm the impacts of dynamic pricing in the spatiotemporal responses of EV traffic flows.

The impact of dynamic CSF scheduling on EV flows can be further evaluated from the traffic flows at FCS2 presented in Fig. 11. It can be seen that FCS2 dynamically adjusts CSF to higher levels to dispatch some EVs to other FCSs for charging, and thereby makes the distribution of EVs more balanced. For example, an excessive number of EVs are observed to be charging during time periods 1 and 5, and a relatively high CSF motivates users to select underutilized FCSs for charging. We further note that dynamic CSF scheduling has only a marginal impact on traffic flows during peak traffic demand periods, such as the time period 3, because all FCSs have reached their full capacity. However, appropriately reducing the CSF can decrease the overall travel costs of users.

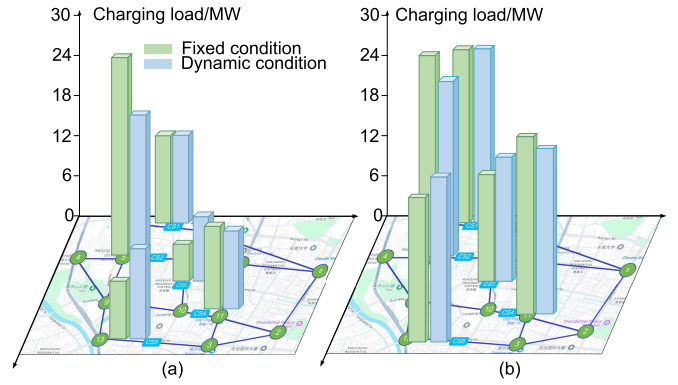


Fig. 12. Charging loads of FCSs under different CSF conditions at different time periods. (a) Time period = 1. (b) Time period = 5.

Accordingly, these results illustrate the sensitivity of dynamic CSF scheduling in the regulation of traffic flow.

The impact of dynamic CSF scheduling on the operating status of the PDN can be evaluated based on the charging loads of FCSs obtained under different CSF conditions presented in Fig. 12 at time periods 1 and 5, respectively. As can be seen, the charging loads of some FCSs are significantly concentrated under the fixed CSF condition during peak hours, which may cause node voltages to exceed the limit or induce local overload conditions. In contrast, the temporal and spatial distributions of loads obtained under dynamic CSF conditions are more balanced, and the peak values are reduced, which helps to alleviate operational pressure on the PDN and enhance system stability. Accordingly, these results confirm the benefits of the spatiotemporal responses of EV charging flows to dynamic CSF scheduling for coupled systems.

E. Economic Analysis of Coupled TN-PDN Systems

To comprehensively evaluate the socioeconomic benefits of the dynamic scheduling strategy, this section conducts a comparative analysis from two dimensions: the travel cost of EV users and the overall operating cost of the coupled system. The travel cost per EV user on the *O-D* pair 1–2 with and without dynamic CSF scheduling can be evaluated over the five time period groups based on the results presented in Fig. 13. In addition, the impacts of the proposed strategy on the various operating costs of the coupled TN-PDN system can be evaluated based on the results listed in Table VI.

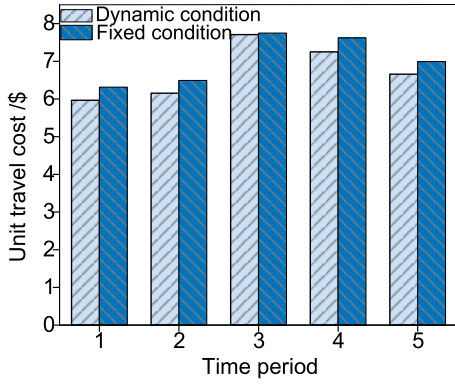


Fig. 13. Unit travel cost on O-D pair 1-2 with and without dynamic CSF scheduling over the five time period groups.

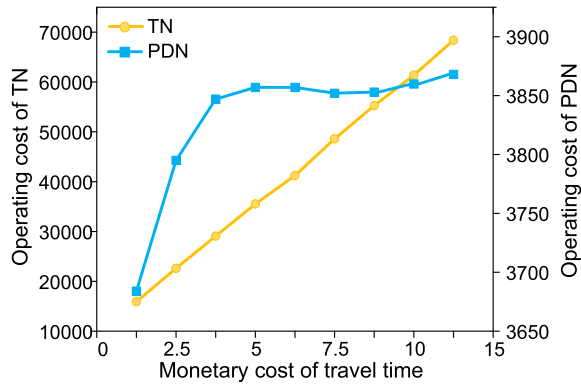


Fig. 14. Sensitivity analysis of the monetary cost of travel time on the operational costs of coupled TN-PDN systems.

As can be seen from Fig. 13, adopting dynamic CSF scheduling reduces the unit travel cost of users over all time periods. For instance, where the traffic demand is relatively low during time period 1, the overall travel time cost for users was reduced even though Fig. 11 indicates that the CSF was increased at this time to alleviate traffic congestion at some charging stations. In fact, some minor benefit was obtained by reducing the CSF under peak traffic demand during period 3, even though charging stations were fully loaded. These results demonstrate the benefit of dynamic CSF scheduling for EV users.

As quantified in Table VI, the proposed strategy yields a 3.5% reduction in total social costs. In the PDN, the scheme achieves a 4.1% operation cost saving and 4.7% power loss reduction by facilitating load leveling across feeders. In the TN, a 3.3% decrease in generalized travel costs is observed, as dynamic pricing effectively steers EVs to less congested stations. This demonstrates that dynamic CSF optimizes the economic performance of the coupled system by simultaneously enhancing grid utilization and traffic equilibrium.

Fig. 14 illustrates the sensitivity of operational costs to the unit passage cost (ω). While TN costs exhibit a linear positive correlation due to the direct weighting of travel time, PDN costs follow a nonlinear trajectory. Specifically, for $\omega < 5\$/h$, PDN cost escalates sharply, as users prioritize time-optimal charging stations, inducing load aggregation that

TABLE VI
OPERATION COSTS OF THE TN AND PDN UNDER DIFFERENT CONDITIONS

	Fixed CSF	Dynamic CSF
DG output (MW)	53.2	50.8
Power purchased from main grid (MW)	12.3	14.7
Operating cost of PDN (\$)	4023.5	3857.1
Net PDN loss (KW)	800.9	763.3
Operating cost of TN (\$)	37239.7	35976.3
Total cost (\$)	41263.3	39833.4

intensifies grid stress. Beyond this threshold, costs saturate, which indicates that physical network constraints (e.g., line capacity and voltage limits) restrict further load redistribution, thereby enforcing a constrained equilibrium despite increasing time values.

V. CONCLUSION

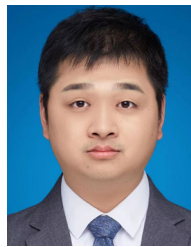
The present work addressed current limitations in the ability to model how charging prices and service fees impact the optimal scheduling of EV charging operations within coupled TN-PDN systems by proposing a graph-attentive surrogate model consisting of a GCN-BCA-transformer architecture to quantify the flexibility of EV charging demands over multiple time periods by learning the nonlinear relationships between charging prices and EV charging responses. The numerical results demonstrated that the proposed model outperforms existing cutting-edge DL methods in predicting the EV charging flow with an average error of 2.26%, while accommodating the issues of uncertain traffic allocation over multiple time periods. Validation on a real dataset in Nanjing indicates the applicability of the model. Further analysis demonstrated that the application of dynamic CSF scheduling effectively alleviates charging station congestion, balances spatiotemporal charging load distributions, while simultaneously reducing the total operating cost of the coupled system by 3.5% and achieving a 4.7% reduction in PDN losses. Therefore, we can conclude that the dynamic scheduling strategy provides economic benefits to both coupled network operators and users.

Furthermore, the present work may assist PDN operators in understanding the price sensitivity of EV charging loads and support TN operators in identifying how spatiotemporal traffic flows influence the flexibility of charging demand. This work currently assumes uniform initial SOC and charging power, which may not fully capture the diversity of real-world demands. Future research will address this by modeling heterogeneous EV behaviors and extending the framework to EV aggregators, exploring their participation in energy markets and ancillary services such as frequency and ramping regulation.

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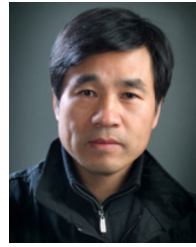


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